

SE212 Database System and Design

Lab 8: DDL and DML (20 points)

Name _____ SID _____

In this lab you will practise the two halves of SQL you have learned in Lecture 7 and 7.1:

- **DDL** (Data Definition Language): `CREATE DATABASE`, `CREATE TABLE`, `DROP`
- **DML** (Data Manipulation Language): `INSERT`, `UPDATE`, `DELETE`, `SELECT`

You need a MySQL server and **MySQL Workbench** installed (see *Before the lab* below). Save every statement you write in one file named `Lab8_<SID>.sql` so you can re-run it.

Before the lab: install the software

You need two things: a **MySQL server** running on your machine, and **MySQL Workbench** to write and run SQL. Pick **Option A** (recommended) or **Option B**. Lab PCs already have the software installed, skip to Part 1.

Option A: MySQL Community Server + MySQL Workbench (recommended)

A1. Download MySQL Community Server. Go to <https://dev.mysql.com/downloads/mysql/>, choose your operating system, and download the installer.

- Windows: choose *Windows (x86, 64-bit)*, *MSI Installer*. Either MSI size works; the larger one works offline.
- macOS: choose the *DMG Archive*. Pick *ARM* for Apple Silicon (M1/M2/M3/M4) or *x86* for Intel Macs.
- On the next page click “**No thanks, just start my download.**” You do not need an Oracle account.

A2. Install the server. Run the installer and accept the defaults (*Typical / Developer Default*). When asked for a **root password**, choose one and **write it down**. Keep *Start MySQL Server at System Startup* ticked. Leave the port at **3306**.

A3. Download and install MySQL Workbench from <https://dev.mysql.com/downloads/workbench/>. Choose your OS, click *Download*, then “*No thanks, just start my download.*” Run the installer with defaults. On Windows, if the installer asks for *Visual C++ Redistributable*, install it from the link it gives and re-run.

A4. Test it. Open MySQL Workbench. Under *MySQL Connections* you should see *Local instance MySQL80* (or similar). Click it, enter your root password, and you should get a query editor. Run:

```
SELECT VERSION();
```

If you get a version number (8.x), you are ready.

Option B: XAMPP (if you already use it, or Option A fails)

B1. Download XAMPP from <https://www.apachefriends.org/download.html>. Choose the version for your OS (any PHP version is fine) and run the installer with defaults. On Windows, install to C:\xampp.

B2. Start the database. Open the *XAMPP Control Panel* and click **Start** next to *MySQL*. It should turn green and show port 3306. You do not need Apache for this lab.

B3. Install MySQL Workbench as in step A3 above. XAMPP ships MariaDB rather than MySQL, so Workbench may show a warning about an unsupported server version. Click *Continue Anyway*.

B4. Connect. In Workbench click + next to *MySQL Connections*, name it `xampp`, host `127.0.0.1`, port 3306, user `root`, **password empty**. Click *Test Connection*, then *OK*.

B5. Command line with XAMPP. The `mysql` client is at C:\xampp\mysql\bin\mysql.exe. In *Command Prompt*, run `cd C:\xampp\mysql\bin` first, then use `mysql -u root` (no `-p`) in the commands in Part 2.

MySQL root password. Every command in this lab logs in as `root`.

- **Your own laptop:** use the password you chose when you installed MySQL Server.
- **Lab PC:** the TA will write the root password on the board: _____
- **XAMPP:** the root password is empty. Leave out `-p`, e.g. `mysql -u root`, and in Workbench leave the password blank.

If none of these work on a lab PC, tell the TA. Do not try to reset the password on a shared machine.

Submission: see the *Submission checklist* on the last page before you upload anything to Mango.

Part 1: University database (12 points)

A small university wants to record students, the courses it offers, the sections of each course that run in a given semester, and which students are enrolled in which sections. The four relations are given below. Underlined attributes are primary keys. $\rightarrow$ shows a foreign key and the table it references.

Relational schema

STUDENT (student_id, first_name, last_name, email, birth_date, major)

Attribute	Type	Constraint
student_id	INT	PRIMARY KEY
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
birth_date	DATE	
major	VARCHAR(50)	

COURSE (course_id, course_name, credits, department)

Attribute	Type	Constraint
course_id	CHAR(6)	PRIMARY KEY
course_name	VARCHAR(100)	NOT NULL
credits	INT	NOT NULL, CHECK (credits BETWEEN 1 AND 4)
department	VARCHAR(50)	NOT NULL

SECTION (section_id, course_id -> COURSE, semester, year, instructor_name, room)

Attribute	Type	Constraint
section_id	INT	PRIMARY KEY, AUTO_INCREMENT
course_id	CHAR(6)	NOT NULL, FK -> COURSE(course_id)
semester	INT	NOT NULL (1 or 2)
year	INT	NOT NULL
instructor_name	VARCHAR(100)	NOT NULL
room	VARCHAR(20)	

ENROLLMENT (student_id -> STUDENT, section_id -> SECTION, enroll_date, grade)

Attribute	Type	Constraint
student_id	INT	FK -> STUDENT(student_id)
section_id	INT	FK -> SECTION(section_id)
enroll_date	DATE	NOT NULL
grade	CHAR(2)	NULL allowed (A, B+, B, C+, C, D+, D, F)
(student_id, section_id)		composite PRIMARY KEY

Tasks

1.1 (3 points) DDL. Write and run the statements to create a database called `se212_university` and the four tables above. Put every constraint from the tables into your `CREATE TABLE` statements (primary keys, foreign keys, NOT NULL, UNIQUE, CHECK). Create the tables in an order that does not break any foreign key.

Hint: start with `DROP DATABASE IF EXISTS se212_university;` so your script can be re-run from the top.

1.2 (4 points) INSERT. Insert the following rows. Insert them in the correct table order.

Note: all student names, IDs, emails and instructor names below are made up for this exercise. They do not belong to real people.

STUDENT

student_id	first_name	last_name	email	birth_date	major
640110001	Nattapong	Srisuwan	nattapong.s@cmu.ac.th	2004-03-12	SE

student_id	first_name	last_name	email	birth_date	major
640110002	Kanokwan	Chaiyasit	kanokwan.c@cmu.ac.th	2003-11-30	SE
640110003	Peeraphat	Wongsa	peeraphat.w@cmu.ac.th	2004-07-08	MMIT

COURSE

course_id	course_name	credits	department
953212	Database System and Database System Design	3	SE
953211	Software Engineering Fundamentals	3	SE

SECTION (let `section_id` auto-increment)

course_id	semester	year	instructor_name	room
953212	1	2026	Pree Thiengburanathum	CAMT 413
953211	1	2026	Somchai Jaidee	CAMT 212

ENROLLMENT

student_id	section_id	enroll_date	grade
640110001	1	2026-06-15	NULL
640110002	1	2026-06-15	NULL
640110003	1	2026-06-16	NULL
640110001	2	2026-06-15	NULL

After inserting, run `SELECT * FROM enrollment;` and take a **screenshot (S1)** of the result grid in Workbench.

1.3 (2 points) UPDATE and DELETE.

- Student 640110001 finished section 1 with grade B+. Write the `UPDATE` statement.
- Student 640110003 dropped section 1. Write the `DELETE` statement that removes only that enrollment.

1.4 (3 points) Short answers. Write your answers below or as comments in your .sql file.

- Why must `COURSE` be created before `SECTION`, and `STUDENT` and `SECTION` before `ENROLLMENT`?
- Try `DELETE FROM section WHERE section_id = 1;` What message does MySQL give and why? Which constraint stops it?
- Try to insert a course with `credits = 5`. What happens and which constraint is responsible?

Take a **screenshot (S2)** of the `se212_university` schema tree in the Workbench Navigator with all four tables expanded.

Part 2: Install the Murach sample databases (8 points)

For the rest of the SQL labs we will use the three sample databases from *Murach's MySQL*: **ap_extra** (accounts payable), **ex** (examples) and **om** (order management). One script creates all of them.

2.1 Get the files

Download Lab8.zip from Mango and extract it. Inside you will find this sheet and a Lab8_murach folder:

```
Lab8/
  Lab8.pdf
  Lab8_murach/
    create_databases.sql          <- creates ap_extra, ex, om
    ER-diagram-APdatabase-Murach.pdf
    ER-diagram-APdatabaseW2ExtraTables-Murach.pdf
    ER-diagram-EXdatabase-Murach.pdf
    ER-diagram-OMdatabase-Murach.pdf
```

2.2 Install from the command line

Step 1. Open a terminal.

- Windows: press Win + R, type cmd, press Enter. If mysql is not recognised, use *MySQL 8.x Command Line Client* from the Start menu instead and skip to Step 3.
- macOS: open *Terminal*. If mysql is not found, run `export PATH=$PATH:/usr/local/mysql/bin` first.

Step 2. Change into the folder where you extracted the zip. Example:

```
cd C:\Users\<you>\Downloads\Lab8\Lab8_murach
```

Step 3. Run the script. Type the command below, then enter the root password you set when installing MySQL.

```
mysql -u root -p < create_databases.sql
```

If you are in the *MySQL Command Line Client* (already logged in), use the `source` command with the full path instead:

```
source C:/Users/<you>/Downloads/Lab8/Lab8_murach/create_databases.sql
```

The script takes a few seconds. No output means success. If you see `ERROR 1045 (28000): Access denied`, the password is wrong; if you see `ERROR 2003 (HY000): Can't connect`, the MySQL server is not running (start it from *Services* or *System Preferences*).

Step 4. Verify. Log in and check the databases exist:

```
mysql -u root -p
SHOW DATABASES;
USE ap_extra;  SHOW TABLES;
USE ex;        SHOW TABLES;
USE om;        SHOW TABLES;
exit
```

You should see 7 tables in `ap_extra`, 11 in `ex`, and 4 in `om`.

Workbench alternative. If the command line does not work: open MySQL Workbench, connect, choose *File > Open SQL Script...*, pick `create_databases.sql`, then click the

lightning-bolt *Execute* button. Right-click in the Navigator and choose *Refresh All* to see the new schemas.

2.3 Tasks

2.1 (2 points). In Workbench, expand the **om** schema and its four tables in the Navigator. Take a **screenshot (S3)**. Open **ER-diagram-OMdatabase-Murach.pdf** and check that the foreign keys in the diagram match the tables.

2.2 (3 points). Using the **ap_extra** database, list the vendor name, city and phone of every vendor located in the state NJ, ordered by vendor name. Write the statement and note how many rows are returned.

Rows returned: _____

2.3 (3 points). Using the **om** database, show each customer's last name together with the number of orders they have placed. Show customers with the most orders first. Write the statement and note the top customer.

Top customer and order count: _____

Submission checklist (read this before you upload)

Submit on **Mango (CMU Canvas)**, assignment **Lab 8**, before the deadline shown there. This is **individual work**. Upload exactly **two files**, named with your student ID:

File 1: Lab8_<SID>.sql

One plain-text SQL script that the TA can run from top to bottom on a clean MySQL server **without any error**. Use this order and these comment headers so the TA can find each part:

```
-- Lab 8 | Name: <your name> | SID: <your SID>
```

```
-- ===== 1.1 DDL =====
```

```
DROP DATABASE IF EXISTS se212_university;
```

```
CREATE DATABASE se212_university;
```

```
USE se212_university;
```

```
-- CREATE TABLE statements here
```

```
-- ===== 1.2 INSERT =====
```

```
-- INSERT statements here
```

```
-- ===== 1.3 UPDATE / DELETE =====
```

```
-- UPDATE and DELETE statements here
```

```
-- ===== 1.4 Short answers =====
```

```
-- 1.4a: <your answer>
```

```
-- 1.4b: <your answer>
```

```
-- 1.4c: <your answer>
```

```
-- ===== 2.2 Vendors in NJ =====
```

```
USE ap_extra;
```

```
-- SELECT here
```

```
-- rows returned: <number>
```

```
-- ===== 2.3 Orders per customer =====
```

```
USE om;
```

```
-- SELECT here
```

```
-- top customer: <last name>, <count> orders
```

File 2: Lab8_<SID>_screenshots.pdf

One PDF (or three PNG files named S1.png, S2.png, S3.png) containing:

Screenshot	What it must show	From task
S1	Workbench result grid of <code>SELECT * FROM enrollment;</code> with 4 rows	1.2
S2	Workbench Navigator: <code>se212_university</code> schema with <code>student</code> , <code>course</code> , <code>section</code> , <code>enrollment</code> expanded	1.4

Screenshot	What it must show	From task
S3	Workbench Navigator: om schema with customers, items, order_details, orders expanded	2.1

Each screenshot must show the **whole Workbench window** (not a cropped grid) so the connection name and schema tree are visible.

How you will be graded

Item	The TA checks	Evidence	Points
1.1	Database + 4 tables created; PK, FK, composite PK, NOT NULL, UNIQUE, CHECK all present; tables in FK-safe order	File 1, section 1.1	3
1.2	All 11 rows inserted, correct order, section_id auto-generated	File 1, section 1.2 + S1	4
1.3	UPDATE and DELETE each affect exactly one row	File 1, section 1.3	2
1.4	Three short answers, correct and explained	File 1, section 1.4	3
2.1	Murach databases installed	S3	2
2.2	Correct SELECT, WHERE, ORDER BY; row count stated	File 1, section 2.2	3
2.3	Correct JOIN, GROUP BY, COUNT, ORDER BY DESC; top customer stated	File 1, section 2.3	3
Total			20

Deductions. Script stops with an error: the TA grades only the parts that ran before the error. Wrong file names or missing SID: -1. Missing screenshot: 0 for that item. Submitted after the deadline: as per course syllabus.